STAT4307_LogLinearV2

Kent Roller

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#Performing draw from population

```
animals<-seq(1:246)
sample1<-sample(animals,50)
sample1
```

```
##  [1] 192   8 146 183 113 230 234 160  67  83 175 133 151  91 141  25 108  95 202
## [20] 150 117   9  56 223 215  33 194  44  79  89 132  77   6 210  74 131   1  29
## [39] 148  43 216  47  24 135 130  16 229  90  85 199
```

```
sample2<-sample(animals,25)
sample2
```

```
##  [1]  15  94 175  86  46   9 114 153 140 105  78  16 196 131 167  56 160 231 228
## [20] 150 207   8 135  30 246
```

```
sample3<-sample(animals,25)
sample3
```

```
##  [1] 212 141  49 180  32  98 211  83 168  22 123 239 121 235 184   9 209 215  92
## [20]  85  37 155  88  48  11
```

#Sorting by draws in common between groups

```
incommon1.2<-intersect(sample1,sample2)
length(incommon1.2)
```

```
## [1] 9
```

```
incommon1.3<-intersect(sample1,sample3)
length(incommon1.3)
```

```
## [1] 5
```

```
incommon2.3<-intersect(sample2,sample3)
length(incommon2.3)
```

```
## [1] 1
```

```
length(intersect(incommon2.3,sample1))
```

```
## [1] 1
```

```
length(setdiff(incommon1.3,sample2))
```

```
## [1] 4
```

```
length(setdiff(incommon1.2,sample3))
```

```
## [1] 8
```

```
length(intersect(setdiff(sample1,sample2),setdiff(sample1,sample3)))
```

```
## [1] 37
```

```
length(setdiff(incommon2.3,sample1))
```

```
## [1] 0
```

```
length(intersect(setdiff(sample3,sample1),setdiff(sample3,sample2)))
```

```
## [1] 20
```

```
length(intersect(setdiff(sample2,sample1),setdiff(sample2,sample3)))
```

```
## [1] 16
```

#Setting up an array with the responses from each sample

```
addiction <- array(data = c(1, 9, 4,1, 8, 3, 38, NA),
                         dim = c(2,2,2),
                         dimnames = list("Sample 1" = c("yes","no"),
                                         "Sample 2" = c("yes","no"),
                                         "Sample 3" = c("yes","no")))
addiction
```

```
## , , Sample 3 = yes
##
##         Sample 2
## Sample 1 yes no
##      yes   1  4
##      no    9  1
##
## , , Sample 3 = no
##
##         Sample 2
## Sample 1 yes no
##      yes   8 38
##      no    3 NA
```

```
ftable(addiction, row.vars = c("Sample 3","Sample 2"))
```

```
##                    Sample 1 yes no
## Sample 3 Sample 2
## yes      yes                  1  9
##          no                   4  1
## no       yes                  8  3
##          no                  38 NA
```

```
addiction.df<- as.data.frame(as.table(addiction))
addiction.df[,-4] <- lapply(addiction.df[,-4], relevel, ref = "no")
addiction.df
```

```
##   Sample.1 Sample.2 Sample.3 Freq
## 1      yes      yes      yes    1
## 2       no      yes      yes    9
## 3      yes       no      yes    4
## 4       no       no      yes    1
## 5      yes      yes       no    8
## 6       no      yes       no    3
## 7      yes       no       no   38
## 8       no       no       no   NA
```

#Creating models using the glm function

```
ci.fit <- glm(Freq ~ Sample.1 + Sample.2 + Sample.3, data = addiction.df, family = poisson)
summary(ci.fit)
```

```
##
## Call:
## glm(formula = Freq ~ Sample.1 + Sample.2 + Sample.3, family = poisson,
##     data = addiction.df)
##
## Deviance Residuals:
##       1        2        3        4        5        6        7
## -0.9109   4.1713  -1.1493  -2.0175  -0.7594  -1.6958   1.0863
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)   3.0604     0.3814   8.024 1.02e-15 ***
## Sample.1yes   0.3956     0.3507   1.128 0.259316
## Sample.2yes  -1.1196     0.2904  -3.856 0.000115 ***
## Sample.3yes  -1.5452     0.3140  -4.921 8.59e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##     Null deviance: 83.699  on 6  degrees of freedom
## Residual deviance: 28.253  on 3  degrees of freedom
##   (1 observation deleted due to missingness)
## AIC: 59.982
##
## Number of Fisher Scoring iterations: 6
```

```
ci.fit1 <- glm(Freq ~ Sample.1 + Sample.2, data = addiction.df, family = poisson)
summary(ci.fit1)
```

```
##
## Call:
## glm(formula = Freq ~ Sample.1 + Sample.2, family = poisson, data = addiction.df)
##
## Deviance Residuals:
##       1        2        3        4        5        6        7
## -3.0051   2.8135  -3.9891  -2.8684   0.1678   0.0202   4.1068
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)   1.9558     0.3032   6.450 1.12e-10 ***
## Sample.1yes   0.9326     0.3137   2.973  0.00295 **
## Sample.2yes  -0.8689     0.2688  -3.233  0.00123 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
```

```
##     Null deviance: 83.699  on 6  degrees of freedom
## Residual deviance: 57.982  on 4  degrees of freedom
##   (1 observation deleted due to missingness)
## AIC: 87.711
##
## Number of Fisher Scoring iterations: 6
```

```
ci.fit2 <- glm(Freq ~ Sample.1, data = addiction.df, family = poisson)
summary(ci.fit2)
```

```
##
## Call:
## glm(formula = Freq ~ Sample.1, family = poisson, data = addiction.df)
##
## Deviance Residuals:
##       1        2        3        4        5        6        7
## -4.2906   1.9552  -2.8681  -1.9324  -1.4292  -0.6785   5.7005
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)   1.4663     0.2774   5.287 1.24e-07 ***
## Sample.1yes   1.0792     0.3107   3.474 0.000514 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##     Null deviance: 83.699  on 6  degrees of freedom
## Residual deviance: 69.191  on 5  degrees of freedom
##   (1 observation deleted due to missingness)
## AIC: 96.92
##
## Number of Fisher Scoring iterations: 5
```

#To get the estimate for the missing cell value:

```r
exp(3.0604)
```

```
## [1] 21.33609
```

```r
#Not needed
exp(1.9558)
```

```
## [1] 7.069572
```

```r
exp(1.466)
```

```
## [1] 4.331873
```

```r
exp(1.2528)
```

```
## [1] 3.50013
```

```r
exp(-1.0986)
```

```
## [1] 0.3333374
```

```r
exp(-0.9268)
```

```
## [1] 0.3958183
```

```r
exp(-0.9268)
```

```
## [1] 0.3958183
```

```r
exp(3.638)
```

```
## [1] 38.01573
```

```r
#The sum of all of the cells that had data:
cell.sum<-sum(c(1, 9, 4,1, 8, 3, 38, 21.33609))
```

```r
#Estimate of population size:
n.hat<-cell.sum+exp(3.0604)
n.hat
```

```
## [1] 106.6722
```

#One list independent of other two:

```
ci.fit3 <- glm(Freq ~ Sample.1 + Sample.2 + Sample.3 + Sample.1*Sample.2, data = addiction.df, family =
summary(ci.fit3)
```

```
##
## Call:
## glm(formula = Freq ~ Sample.1 + Sample.2 + Sample.3 + Sample.1 *
##     Sample.2, family = poisson, data = addiction.df)
##
## Deviance Residuals:
##       1        2        3        4        5        6        7
## -0.7834   3.0378  -1.9719   0.0000   0.3695  -2.4201   0.9093
##
## Coefficients:
##                         Estimate Std. Error z value Pr(>|z|)
## (Intercept)               1.2528     1.0449   1.199   0.2306
## Sample.1yes               2.2336     1.0389   2.150   0.0316 *
## Sample.2yes               0.9808     1.0672   0.919   0.3581
## Sample.3yes              -1.2528     0.3030  -4.134 3.57e-05 ***
## Sample.1yes:Sample.2yes  -2.5213     1.1286  -2.234   0.0255 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##     Null deviance: 83.699  on 6  degrees of freedom
## Residual deviance: 20.551  on 2  degrees of freedom
##   (1 observation deleted due to missingness)
## AIC: 54.28
##
## Number of Fisher Scoring iterations: 5
```

```
ci.fit6 <- glm(Freq ~ Sample.1 + Sample.2 + Sample.3 + Sample.1*Sample.3 , data = addiction.df, family =
summary(ci.fit6)
```

```
##
## Call:
## glm(formula = Freq ~ Sample.1 + Sample.2 + Sample.3 + Sample.1 *
##     Sample.3, family = poisson, data = addiction.df)
##
## Deviance Residuals:
##       1        2        3        4        5        6        7
## -0.4159   2.8239   0.2478  -2.8623  -1.6396   0.0000   0.9526
##
## Coefficients:
##                         Estimate Std. Error z value Pr(>|z|)
## (Intercept)              1.96944    0.64198   3.068  0.00216 **
## Sample.1yes              1.50953    0.62788   2.404  0.01621 *
## Sample.2yes             -0.87083    0.28073  -3.102  0.00192 **
## Sample.3yes             -0.01653    0.68738  -0.024  0.98082
## Sample.1yes:Sample.3yes -2.20267    0.83321  -2.644  0.00820 **
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##     Null deviance: 83.699  on 6  degrees of freedom
## Residual deviance: 19.997  on 2  degrees of freedom
##   (1 observation deleted due to missingness)
## AIC: 53.726
##
## Number of Fisher Scoring iterations: 6
```

```
ci.fit7 <- glm(Freq ~ Sample.1 + Sample.2 + Sample.3 + Sample.2*Sample.3, data = addiction.df, family =
summary(ci.fit7)
```

```
##
## Call:
## glm(formula = Freq ~ Sample.1 + Sample.2 + Sample.3 + Sample.2 *
##     Sample.3, family = poisson, data = addiction.df)
##
## Deviance Residuals:
##       1        2        3        4        5        6        7
## -2.1866   1.6063   0.8718  -1.0805   0.9975  -1.1676   0.0000
##
## Coefficients:
##                            Estimate Std. Error z value Pr(>|z|)
## (Intercept)               3.638e+00  4.245e-01   8.570  < 2e-16 ***
## Sample.1yes               5.976e-11  3.922e-01   0.000    1e+00
## Sample.2yes              -1.933e+00  3.946e-01  -4.899 9.65e-07 ***
## Sample.3yes              -2.721e+00  5.146e-01  -5.289 1.23e-07 ***
## Sample.2yes:Sample.3yes   2.626e+00  6.750e-01   3.890    1e-04 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##     Null deviance: 83.699  on 6  degrees of freedom
## Residual deviance: 11.647  on 2  degrees of freedom
##   (1 observation deleted due to missingness)
## AIC: 45.376
##
## Number of Fisher Scoring iterations: 5
```

#To get the estimate for the missing cell value:

```
exp(1.2528)
```

```
## [1] 3.50013
```

```
exp(1.96944)
```

```
## [1] 7.166662
```

```r
exp(3.638)
```

```
## [1] 38.01573
```

#Estimate of population size:

```r
n.hat1<-cell.sum+exp(1.2528)
n.hat1
```

```
## [1] 88.83622
```

```r
n.hat2<-cell.sum+exp(1.96944)
n.hat2
```

```
## [1] 92.50275
```

```r
n.hat3<-cell.sum+exp(3.638)
n.hat3
```

```
## [1] 123.3518
```

#Two independent given the third:

```
ci.fit4 <- glm(Freq ~ Sample.1 + Sample.2 + Sample.3+ Sample.1*Sample.3+Sample.1*Sample.2, data = addic
summary(ci.fit4)
```

```
##
## Call:
## glm(formula = Freq ~ Sample.1 + Sample.2 + Sample.3 + Sample.1 *
##     Sample.3 + Sample.1 * Sample.2, family = poisson, data = addiction.df)
##
## Deviance Residuals:
##        1         2         3         4         5         6         7
##  0.12261   0.00000  -0.05826   0.00000  -0.04139   0.00000   0.01911
##
## Coefficients:
##                         Estimate Std. Error z value Pr(>|z|)
## (Intercept)              -1.0986     1.2018  -0.914 0.360662
## Sample.1yes               4.7331     1.2126   3.903 9.49e-05 ***
## Sample.2yes               2.1972     1.0541   2.084 0.037117 *
## Sample.3yes               1.0986     0.6667   1.648 0.099369 .
## Sample.1yes:Sample.3yes  -3.3178     0.8162  -4.065 4.80e-05 ***
## Sample.1yes:Sample.2yes  -3.7377     1.1163  -3.348 0.000813 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##     Null deviance: 83.699148  on 6  degrees of freedom
## Residual deviance:  0.020504  on 1  degrees of freedom
##   (1 observation deleted due to missingness)
## AIC: 35.75
##
## Number of Fisher Scoring iterations: 3
```

```
ci.fit8 <- glm(Freq ~ Sample.1 + Sample.2 + Sample.3+ Sample.2*Sample.3+Sample.1*Sample.2, data = addic
summary(ci.fit8)
```

```
##
## Call:
## glm(formula = Freq ~ Sample.1 + Sample.2 + Sample.3 + Sample.2 *
##     Sample.3 + Sample.1 * Sample.2, family = poisson, data = addiction.df)
##
## Deviance Residuals:
##      1       2       3       4       5       6       7
## -1.913   1.267   0.000   0.000   1.375  -1.461   0.000
##
## Coefficients:
##                         Estimate Std. Error z value Pr(>|z|)
## (Intercept)               2.2513     1.1297   1.993  0.04629 *
## Sample.1yes               1.3863     1.1180   1.240  0.21500
## Sample.2yes              -0.4130     1.1845  -0.349  0.72732
## Sample.3yes              -2.2513     0.5257  -4.283 1.85e-05 ***
## Sample.2yes:Sample.3yes   2.1560     0.6835   3.154  0.00161 **
```

```
## Sample.1yes:Sample.2yes  -1.6740     1.2019  -1.393  0.16367
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##     Null deviance: 83.6991  on 6  degrees of freedom
## Residual deviance:  9.2895  on 1  degrees of freedom
##   (1 observation deleted due to missingness)
## AIC: 45.019
##
## Number of Fisher Scoring iterations: 5
```

```
ci.fit9 <- glm(Freq ~ Sample.1 + Sample.2 + Sample.3+ Sample.2*Sample.3+Sample.1*Sample.3, data = addic
summary(ci.fit9)
```

```
##
## Call:
## glm(formula = Freq ~ Sample.1 + Sample.2 + Sample.3 + Sample.2 *
##     Sample.3 + Sample.1 * Sample.3, family = poisson, data = addiction.df)
##
## Deviance Residuals:
##       1        2        3        4        5        6        7
## -1.5029   0.8575   1.5288  -1.5029   0.0000   0.0000   0.0000
##
## Coefficients:
##                          Estimate Std. Error z value Pr(>|z|)
## (Intercept)                2.6568     0.6962   3.816 0.000135 ***
## Sample.1yes                0.9808     0.6770   1.449 0.147399
## Sample.2yes               -1.5581     0.3890  -4.006 6.19e-05 ***
## Sample.3yes               -1.4528     0.8473  -1.715 0.086432 .
## Sample.2yes:Sample.3yes    2.2513     0.6718   3.351 0.000805 ***
## Sample.1yes:Sample.3yes   -1.6740     0.8708  -1.922 0.054569 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##     Null deviance: 83.6991  on 6  degrees of freedom
## Residual deviance:  7.5897  on 1  degrees of freedom
##   (1 observation deleted due to missingness)
## AIC: 43.319
##
## Number of Fisher Scoring iterations: 5
```

#To get the estimate for the missing cell value:

```
exp(-1.0986)
```

```
## [1] 0.3333374
```

```r
exp(2.2513)
```

```
## [1] 9.500078
```

```r
exp(2.6568)
```

```
## [1] 14.25061
```

#Estimate of population size:

```r
n.hat4<-cell.sum+exp(-1.0986)
n.hat4
```

```
## [1] 85.66943
```

```r
n.hat5<-cell.sum+exp(2.2513)
n.hat5
```

```
## [1] 94.83617
```

```r
n.hat6<-cell.sum+exp(2.6568)
n.hat6
```

```
## [1] 99.5867
```

#All two way interactions:

```
ci.fit5 <- glm(Freq ~ Sample.1 + Sample.2 + Sample.3 +Sample.1*Sample.3+Sample.1*Sample.2+Sample.2*Sampl

summary(ci.fit5)
```

```
##
## Call:
## glm(formula = Freq ~ Sample.1 + Sample.2 + Sample.3 + Sample.1 *
##     Sample.3 + Sample.1 * Sample.2 + Sample.2 * Sample.3, family = poisson,
##     data = addiction.df)
##
## Deviance Residuals:
## [1]  0  0  0  0  0  0  0
##
## Coefficients:
##                           Estimate Std. Error z value Pr(>|z|)
## (Intercept)                -0.9268     1.6869  -0.549  0.58275
## Sample.1yes                 4.5643     1.6791   2.718  0.00656 **
## Sample.2yes                 2.0254     1.5851   1.278  0.20132
## Sample.3yes                 0.9268     1.3586   0.682  0.49514
## Sample.1yes:Sample.3yes    -3.1781     1.2528  -2.537  0.01119 *
## Sample.1yes:Sample.2yes    -3.5835     1.5366  -2.332  0.01969 *
## Sample.2yes:Sample.3yes     0.1719     1.1838   0.145  0.88457
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##     Null deviance: 8.3699e+01  on 6  degrees of freedom
## Residual deviance: 4.6629e-15  on 0  degrees of freedom
##   (1 observation deleted due to missingness)
## AIC: 37.729
##
## Number of Fisher Scoring iterations: 3
```

#To get the estimate for the missing cell value:

```
exp(-0.9268)
```

```
## [1] 0.3958183
```

#Estimate of population size:

```
n.hat7<-cell.sum+exp(-0.9268)
n.hat7
```

```
## [1] 85.73191
```